

MINOR PROJECT 1

Project Title:

Predicting Indian Premier League (IPL) Match Outcomes Using Random Forest

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TITLE

Predicting IPL Match Outcomes Using Random Forest.

PROBLEM STATEMENT

Predicting the winning team of an IPL cricket match based on historical match data, including venue, toss decisions, and competing teams.

DATASET DESCRIPTION

The dataset utilized for this project is the "**IPL Complete Dataset**" sourced from Kaggle. It contains historical data of IPL matches spanning multiple seasons.

- **Dataset Link:** <https://www.kaggle.com/datasets/patrickb1912/ipl-complete-dataset-20082020>
- **Target Variable:** winner
- **Features Used:** team1, team2, toss_winner, toss_decision, and venue.

DATA PREPROCESSING STEPS

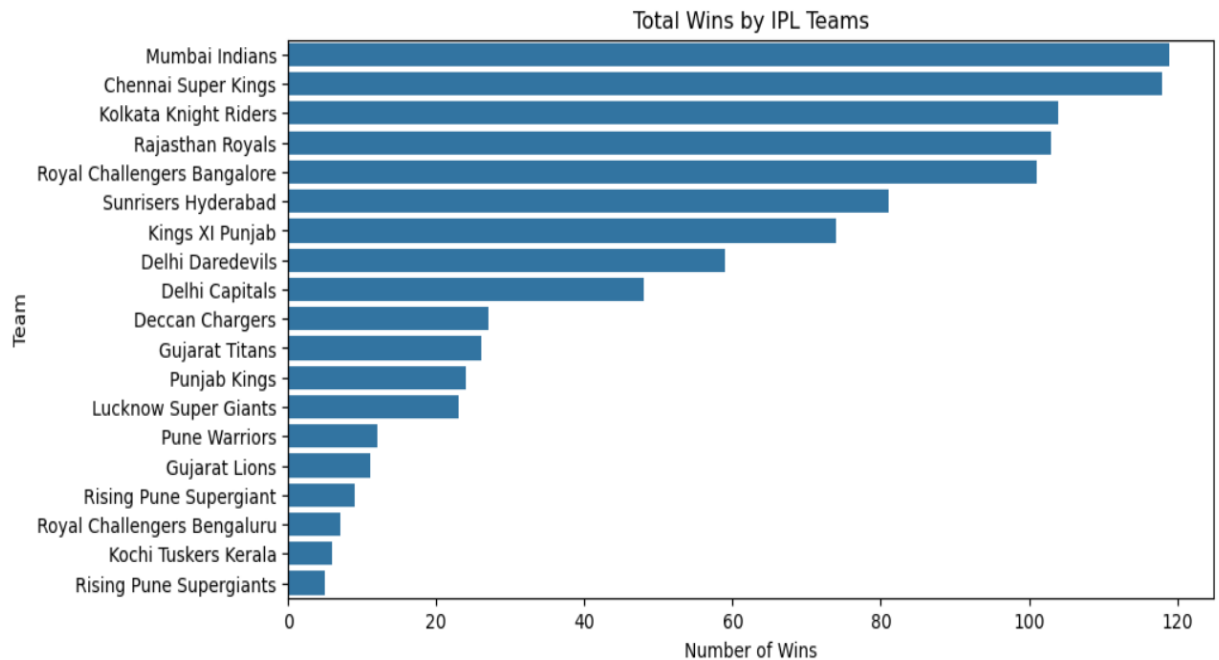
To prepare the raw data for the machine learning model, several preprocessing steps were executed:

- **Duplicate Removal:** Checked for and removed any duplicate rows to prevent model bias.
- **Feature Selection:** Dropped irrelevant columns (like umpires or match dates) to focus strictly on the teams, toss dynamics, and venue.

- **Handling Missing Values:** Matches that ended with no result (e.g., due to rain) resulted in missing values in the `winner` column. These specific rows were dropped.
- **Feature Encoding:** Machine learning models require numerical input. `LabelEncoder` from Scikit-Learn was used to convert categorical string data (like team names and venues) into unique integer values.
- **Feature Scaling:** `StandardScaler` was applied to normalize the feature variables, ensuring that no single feature disproportionately influenced the model.

EXPLORATORY DATA ANALYSIS (EDA)

During the Exploratory Data Analysis phase, a count plot was generated to visualize the total number of historical match wins per team.



- **Analysis:** The visualization highlights the historical dominance of specific franchises across the seasons, such as Chennai Super Kings and Mumbai Indians. This distribution helps baseline the dataset and identify class imbalances in the target variable.

MODEL USED

The model selected for this project is the **Random Forest Classifier**. This specific algorithm was chosen because it is highly effective at handling non-linear data and complex categorical features (like team names and venues). By building an ensemble of multiple decision trees, Random Forest reduces the risk of overfitting compared to a single decision tree model.

TRAINING PROCESS

The dataset was split using the `train_test_split` method to evaluate the model on unseen data.

- **Training Set:** 80% of the dataset was used to train the model.
- **Testing Set:** 20% of the dataset was reserved for evaluating model performance.
- **Hyperparameters:** The Random Forest model was initialized with `n_estimators=100` (creating 100 internal decision trees) and a `random_state=42` to ensure the results are reproducible.

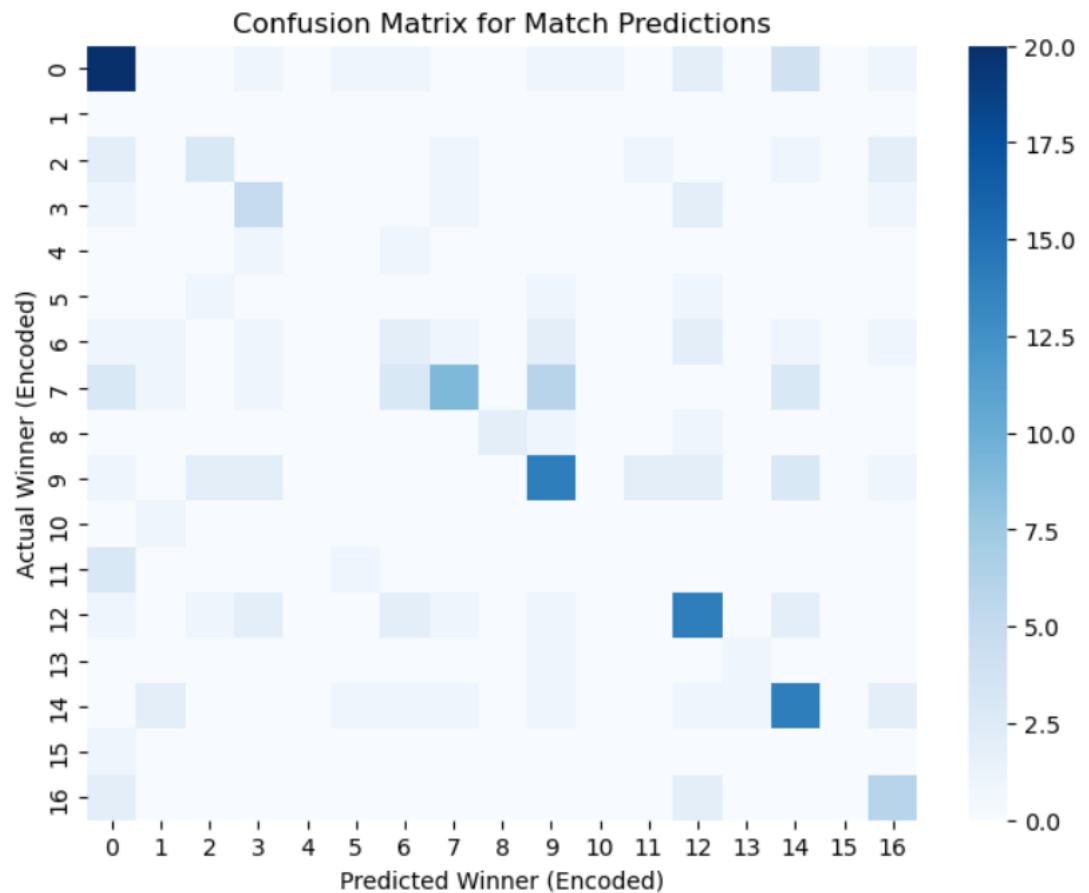
EVALUATION METRICS AND RESULTS

The model was evaluated using standard classification metrics.

- **Accuracy:** 0.4688 - This represents the overall percentage of matches the model predicted correctly.
- **Precision:** 0.4831 - This indicates the accuracy of the positive predictions.
- **Recall:** 0.4688 - This measures the model's ability to find all the correct winning instances.
- **F1-Score:** 0.4655 - The harmonic mean of precision and recall.

Confusion Matrix Analysis:

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- The confusion matrix visualizes the exact breakdown of true positive predictions versus false classifications across all encoded IPL teams, showing exactly where the model succeeded and where it confused one team for another.

CONCLUSION

This minor project successfully demonstrates a complete machine learning workflow. By utilizing a Random Forest Classifier, we were able to predict IPL match outcomes based on toss decisions and venue locations. The evaluation metrics indicate that while cricket is an inherently unpredictable game, historical data regarding toss

advantages and venue familiarity provides statistically significant signals for predicting the winning team.